

LoRaWAN-as-a-Service for Rwanda — Business Concept

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Executive Summary

We are building **the first private LoRaWAN connectivity-and-solutions network for Rwanda**. The network provides long-range, low-power, low-cost wireless communication for IoT devices — purpose-built for the use cases that matter to Rwandan utilities, agriculture, logistics, and smart-city projects.

Our offering bundles three things into a single recurring revenue model:

1. **Network coverage** — gateways we deploy across Kigali (Phase 1) and secondary cities (Phase 2-3) that any compatible LoRaWAN sensor can talk to
2. **Solution stack** — dashboards, payload decoders, integrations and APIs that turn raw sensor data into business decisions
3. **Lifecycle services** — installation, maintenance, training and SLA-backed support tailored to the Rwandan context

The infrastructure is already operational at proof-of-concept stage: a working ChirpStack v4 network server, one gateway online, full software stack tested end-to-end. We need 6-12 months of seed funding to convert the proof of concept into Rwanda's first commercial LoRaWAN operator.

Why this matters now: Rwanda's water utility (WASAC) loses ~38% of produced water — roughly **Rwf 9.9 billion (~USD 7.5M) annually** — to leaks, faulty meters, and unbilled consumption. Rwanda's Strategic Plan for Agricultural Transformation (PSTA5) targets 83,250 hectares for climate-smart upgrades benefiting 170,200 farmers and 375 agribusinesses. Both sectors urgently need affordable, scalable IoT — and neither has an in-country wholesale LoRaWAN provider serving them today.

1. The Problem

1.1 What Rwanda actually loses to "no IoT connectivity"

Sector	Quantified annual loss	Source
Water (NRW losses, WASAC)	~Rwf 9.9 B (~USD 7.5M)	RURA / WASAC public reports
Post-harvest agricultural losses	15-40% of production	MINAGRI Strategic Plan PSTA5
Cold-chain failures (vaccines, dairy, horticulture exports)	Not consolidated; significant	Industry estimates
Electricity meter reading inefficiency	Manual, monthly	REG operational reports
Asset/vehicle theft and route inefficiency	Significant; uncounted	Logistics industry estimates

1.2 Why current solutions don't fit

Three connectivity options exist for IoT in Rwanda today; **none** is right for high-volume, low-cost sensor deployments:

Technology	Issue for Rwandan IoT use cases
Cellular (3G/4G via MTN, Airtel)	\$0.50-2/device/month airtime, plus SIM, plus per-MB data costs. Power-hungry (~weeks of battery, not years). Coverage gaps in rural areas.
WiFi/Ethernet	Requires existing site network. Unworkable for distributed assets (water meters across a city, sensors across a farm).
Satellite (Starlink, etc.)	Cost-prohibitive per endpoint. Power-hungry. Not designed for cheap sensors.
LoRaWAN (the gap we fill)	\$0.50/device/month total connectivity. 5-10 year battery life. 2-15 km range per gateway. No public LoRaWAN operator exists in Rwanda today.

1.3 The strategic timing

- **MWC Kigali** has positioned Rwanda as Africa's connectivity proving ground
- **MINAGRI's Climate-Smart Agriculture Investment Plan** (launched June 2025 with IFC) explicitly targets digital agriculture

- **WASAC Group** is publicly soliciting private-sector partnerships for smart metering at scale
- **City of Kigali Master Plan 2050** includes IoT-based water, lighting, waste management
- **Anthropic-Rwanda MoU** signed in 2025 signals government openness to digital infrastructure investment

There is a short window — 18 to 36 months — to establish Rwanda's first LoRaWAN network operator before a foreign player does. Once a competitor lays gateways across Kigali, the moat is theirs.

2. The Solution

2.1 What we operate

A **multi-tenant private LoRaWAN network** running on ChirpStack v4 (open-source, Apache 2.0 — no licensing fees, full commercial freedom):

- Gateways: a mix of indoor and outdoor units across Kigali and progressively expanding
- Network server: hosted (initially on Rwandan cloud, later sovereign deployment)
- MQTT/REST integration APIs: customers receive their data as JSON, with the protocol details abstracted away
- Customer-facing dashboards: built per vertical (water, agriculture, logistics, etc.) on Grafana / ThingsBoard / custom

2.2 What customers actually buy from us

We offer three product tiers — chosen because they map to how customers currently buy IT services:

Tier 1 — Connectivity-as-a-Service ("NaaS")

- Customer brings their own LoRaWAN sensors and dashboards
- We provide network coverage and a tenant in our ChirpStack
- Pricing: **USD 0.50–3.00 per device per month**, volume-tiered
- Margin: ~85% gross after gateway amortisation
- Target customer: technical integrators, OEMs, donor-funded projects with hardware

Tier 2 — Vertical Solution Bundle ("Outcome-as-a-Service")

- We deliver sensors + connectivity + dashboard + support as a single subscription
- Examples: water tank monitoring, soil moisture for greenhouses, asset tracking, cold-chain

- Pricing: **USD 5-25 per asset per month** depending on hardware cost amortisation
- Margin: 40-60% — lower than NaaS but order-of-magnitude larger market
- Target customer: utilities, cooperatives, hospitals, NGOs

Tier 3 — Coverage-as-a-Service ("Wholesale roaming")

- Other IoT operators or international networks (TTN, Helium, Senet) buy access to our coverage for their devices roaming in Rwanda
- Pricing: per-message wholesale
- Target: deferred to Year 3+ when we have 50+ gateways

2.3 Technical proof of concept (already built)

What we have running today (as of June 2026):

-  ChirpStack v4 fully deployed and operational
-  One gateway (Seeed WM1302 SPI on Raspberry Pi 4) commissioned and forwarding packets
-  MQTT and REST APIs working end-to-end
-  Documentation, runbooks, deployment record (this is rare among Rwandan tech ventures and signals operational seriousness)
-  Procurement in progress: production-grade EU868 hardware (gateways and end-devices)
-  RURA Type Approval application in preparation

2.4 Why ChirpStack (vs The Things Stack / commercial alternatives)

We deliberately chose ChirpStack over The Things Industries' stack because:

- Apache 2.0 — zero per-device licensing fees forever
 - Full commercial use without negotiating with a foreign vendor
 - Identical LoRaWAN protocol compatibility — devices and gateways from any vendor work
 - Maintained by Semtech (the chip company), strong long-term outlook
 - Data sovereignty: the entire network can run on Rwandan-hosted infrastructure if customers require it
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3. Market Sizing

3.1 Bottom-up — addressable revenue in Rwanda by 2030

Vertical	Endpoints by 2030 (est.)	ARPU/mo (USD)	Annual revenue potential
Water meters (WASAC + private wells)	250,000	1.50	USD 4.5M
Smart agriculture (farms + cooperatives)	100,000	2.50	USD 3.0M
Cold-chain / pharma	20,000	8.00	USD 1.9M
Asset / vehicle tracking	50,000	4.00	USD 2.4M
Smart city (lighting, waste, parking)	30,000	3.00	USD 1.1M
Industrial / mining	15,000	5.00	USD 0.9M
Total addressable (Rwanda alone)	465,000 endpoints	—	~USD 13.8M / year

3.2 Regional expansion

The same network architecture (and same software stack) can replicate across the East African Community:

- Burundi (REGIDESO is also working with Itron on water metering — similar profile to WASAC)
- Tanzania (large agriculture sector, large geography)
- Uganda (similar profile + large logistics corridor traffic)
- Kenya (already has nascent LoRaWAN ecosystem — competitive but bigger pie)

Adding regional markets pushes 2030 addressable revenue to **~USD 50M+** without changing the platform.

3.3 Comparable players outside Rwanda

Operator	Country	Status	Lesson
Upande	Kenya	Profitable, water	Their pivot to single vertical (water) is what made

Operator	Country	Status	Lesson
Limited		utility focus	them sustainable. Mirror this.
Sigfox South Africa	South Africa	Shut down 2022	Wrong tech (Sigfox is proprietary and discontinued). LoRaWAN's open standard is structurally superior.
TagoIO / Datacake	International	Platform-only	Confirms there's appetite — but they need network operators to bring data. We are that operator.

4. Target Customers — Who to Approach

This is the operational list. Companies are tagged with: **why they fit, realistic deal size, best entry door, and expected sales cycle.**

4.1 Tier-A — Government & utilities (high deal size, slow cycle)

#	Organisation	Why they need us	Realistic Year-1 contract	Best entry door	Cycle
1	WASAC Group (Water and Sanitation Corporation)	Publicly soliciting private-sector partnerships on smart metering. Has 160,000+ Itron meters but no IoT network layer connecting them to a single platform. NRW reduction is their flagship KPI.	USD 50-200k pilot (10,000 meters, 1 year)	Tender / RFP route. Also direct meeting via WASAC CEO Dr. Kabasha or via the Itron partnership team.	6-18 months
2	REG (Rwanda Energy Group) / EUCL	Manual electricity meter reading is the obvious sister problem. Smart electricity grid roadmap referenced in MINICT plans.	USD 30-100k pilot	RFP + direct via REG innovation team	12-24 months
3	City of Kigali (CoK)	Smart city ambitions, traffic, lighting, waste, parking. Mayor's office has innovation mandate.	USD 30-80k street-lighting pilot	Direct via CoK Innovation Office	9-18 months

#	Organisation	Why they need us	Realistic Year-1 contract	Best entry door	Cycle
4	Ministry of Agriculture (MINAGRI)	Owns the Climate-Smart Agriculture Investment Plan. Direct buyer of demonstration projects.	USD 50-150k pilot via PSTA5 budget	Direct via the Director General responsible for Digital Agriculture	6-12 months
5	Rwanda Development Board (RDB)	Gatekeeper for any project of size; provides incentives and one-stop facilitation. Not a buyer but an enabler.	(Enabler, not customer)	Direct meeting via Investment Promotion Department	n/a
6	RURA (Rwanda Utilities Regulatory Authority)	Issues type approval; sets spectrum policy. Must be onboard.	(Regulator)	Direct submission of type approval application + briefing	2-3 months for approval

4.2 Tier-B — Commercial agriculture & agribusiness (mid deal size, faster cycle)

#	Organisation	Why they fit	Realistic Year-1 contract	Best entry door
7	Inyange Industries	Largest dairy and beverage processor. Cold-chain monitoring across collection centres.	USD 20-60k for fleet of 50-200 sensors	Direct via Operations Director
8	Bakhresa Grain Milling Rwanda	Grain silos, warehouses. Temperature/humidity monitoring prevents spoilage.	USD 15-40k	Direct via Plant Manager
9	Rwanda Mountain Tea / Sorwathe	Tea factories, plantation conditions, drying houses. Quality control critical for export pricing.	USD 25-75k	Direct via Operations / Sustainability Manager
10	NAEB (National Agricultural Export Development Board)	Coffee, tea, horticulture export quality requires traceability. Donor-funded budget for tech upgrades.	USD 30-100k for traceability pilot	NAEB Innovation team
11	Coffee cooperatives via NAEB / IWCA	800+ washing stations need temperature, water-use,	USD 5-15k per cooperative;	Via NAEB / cooperative

#	Organisation	Why they fit	Realistic Year-1 contract	Best entry door
	Rwanda	fermentation monitoring	volume play	federations
12	Holland Greentech Rwanda / Smart Farms	Greenhouse operators already buying IoT. Easiest "Tier 1" customers — they have devices, need network.	USD 5-15k connectivity-only	Direct
13	Kumva Insights (existing player)	Already doing IoT for farms. Could be a customer (use our network) OR a partner.	Partnership, not contract	Direct CEO meeting

4.3 Tier-C — Health, logistics, hospitality (smaller, fastest-closing)

#	Organisation	Use case	Realistic Year-1 contract
14	University Teaching Hospitals (CHUK, CHUB)	Cold-chain (vaccines, blood, samples)	USD 10-30k
15	Pharmacie du 1000 Collines / large pharmacy chains	Cold-chain across branches	USD 5-15k
16	Akagera Aviation / Bralirwa logistics fleets	Vehicle tracking, fuel monitoring	USD 10-30k
17	DHL Rwanda, Bolloré Africa Logistics	Asset tracking for transit goods	USD 20-50k
18	Kigali Marriott, Serena, Radisson	Energy monitoring, water tanks, kitchen cold storage	USD 5-15k each
19	Volkswagen Mobility Solutions Rwanda	Mobility data, fleet telematics if they push beyond ride-share	USD 15-40k

4.4 Tier-D — International donors and NGOs (grant-funded buyers, can be very fast)

#	Organisation	Why they fund this	Realistic engagement
20	Water For People Rwanda	Already funding rural water metering kits with USAID money. Natural fit for our network.	Sub-contract on existing project; USD 20-80k
21	GIZ Rwanda	German development cooperation; large climate adaptation portfolio	Implementing partner role; USD 50-200k
22	JICA Rwanda	Funded the WASAC NRW Reduction Project (2018-2020). Likely to fund follow-on.	Partner / sub-contractor
23	USAID Rwanda — Feed the Future	Agriculture digital tools fit their mandate	USD 50-300k via implementing partner
24	GGGI (Global Green Growth Institute)	Has Rwanda office. Climate-smart agriculture focus.	Project partnership
25	One Acre Fund (Tubura Rwanda)	Reaches 200k+ Rwandan smallholders. Could deploy our network for their farmer data needs	Strategic partnership / paid pilot

4.5 Tier-E — Tech ecosystem partners (not buyers, but multipliers)

#	Organisation	Why engage
26	Smart Africa (Kigali HQ)	Continental policy body. Endorsement = continental credibility.
27	Kigali Innovation City	Co-location for offices + visibility
28	NCST (National Council for Science and Technology)	Research collaboration grants (see Section 7)
29	ACE-IoT (African Center of Excellence in IoT) at UR	Academic partner for talent pipeline + research credibility
30	Norrsken Kigali / Westerwelle Startup Haus / Impact Hub Kigali	Community, mentors, demo days, investor connections
31	MTN Rwanda / Airtel Rwanda	NOT competitors — they sell cellular. Potential partners for backhaul connectivity to remote gateways and bundled

#	Organisation	Why engage
		offerings.

5. Business Model

5.1 Revenue mix (3-year target)

Year 1 (validation):	NaaS 30% Solutions 60% Pilots/grants 10%
Year 2 (scale):	NaaS 40% Solutions 50% Pilots 10%
Year 3 (consolidation):	NaaS 35% Solutions 55% Wholesale 10%

5.2 Unit economics (illustrative; pilot phase)

Single 1,000-device water utility pilot:

- One-time hardware: 1,000 meters × \$80 = \$80,000 (financed by customer)
- One-time installation: \$10,000
- Connectivity: 1,000 × \$1.50/mo × 12 = \$18,000/year (our recurring)
- Software/dashboards: \$6,000/year (our recurring)
- Support/SLA: \$4,000/year (our recurring)
- **Our recurring per customer: USD 28k/year**
- Gross margin on connectivity: ~85%
- Customer ROI on NRW recovered: water utility recovers ~30% of lost revenue → ~\$200k/year savings → 7-month payback

5.3 Cost structure (Year 1)

Category	USD/year
Salaries (3-4 people)	60,000
Cloud hosting (Hetzner / local)	1,200
Gateway hardware (20 units for Phase 1)	30,000
RURA type approval + legal	8,000
Office (Norrskén/Innovation Village)	6,000
Marketing, travel, events	10,000

Category	USD/year
Reserves & misc	15,000
Total	~130,000

6. Competitive Landscape

Competitor type	Player	Threat level	Response
Direct LoRaWAN operator in Rwanda	None today	—	First-mover advantage — must move fast
LoRaWAN operators in neighbouring countries (Kenya: Upande)	Could enter Rwanda	Medium (24-36 months)	Lock in WASAC and one anchor agriculture client early
Cellular IoT (NB-IoT via MTN)	MTN Rwanda	Medium for some verticals	Differentiate on cost-per-device-per-year; partner where useful
Foreign hardware vendors selling direct (Itron, Kamstrup)	Itron at WASAC	High in their installed base	Position as the IoT/data layer on top of their meters
Existing IoT consultancies (Kumva Insights, Agri-IoT)	Local players	Low-medium	Partner: we provide network, they provide vertical solutions
TTN community network	None in Rwanda	Low	If activated, complementary (we run commercial, they run hobbyist)

7. Grant Application Section

This is the most concrete part of this document for funder conversations. We are eligible for **multiple parallel funding programs**, each with different criteria and decision timelines. Apply in parallel — not sequentially.

7.1 Local grants (Rwanda)

Programme	Funder	Amount	Best fit	Application window
Hanga Pitchfest	MINICT + RDB + UNDP	USD 10-50k (cash + in-kind)	Early-stage proof of traction. We qualify now.	Annual, typically Q3-Q4
Innovate Rwanda Platform	Government of Rwanda	Variable, ecosystem support	Visibility, networking, sometimes grants	Rolling
Rwanda Innovation Fund	Managed by Angaza Capital, sized USD 30M	USD 100k-1M equity / convertible	Once we have a paying pilot. ~Month 9-12.	Rolling
NCST Collaborative R&I Grants	National Council for Science and Technology	USD 20-100k	If we pair with a university (ACE-IoT or UR) — manufacturing or smart-ag focus	Annual calls
Rwanda Digital Acceleration Project (RDAP) — Early Stage Financing	World Bank-funded, MINICT executing	USD 10-100k	We qualify based on digital + climate + inclusion narrative	Rolling, currently active
Rwanda Green Fund (FONERWA)	Government, climate finance	USD 50-500k	If we frame as climate adaptation (water-saving, climate-smart ag)	Quarterly review

7.2 International grants (most relevant)

Programme	Funder	Amount	Best fit	Window
GSMA Innovation Fund for Green Transition for Mobile	GSMA	GBP 100k-200k (~USD 130-260k)	Excellent fit: digital + climate + LMIC + mobile/IoT	Recent rounds close April annually — confirm next round

Programme	Funder	Amount	Best fit	Window
GSMA Innovation Fund for Digital Utilities	GSMA	GBP 100-250k (~USD 130-330k)	Our exact use case (water utilities)	Rolling rounds; check active calls
AfDB/ Africa Digital Financial Inclusion Facility (ADFI)	African Development Bank	Up to USD 1M	If we connect connectivity to financial inclusion (pre-paid water meters as a financial-inclusion play)	Rolling
AIIB Project Preparation Special Fund	Asian Infrastructure Investment Bank	Project-specific	Already funding the Kigali Climate-Smart Wastewater project — natural extension to IoT layer	Via WASAC as primary applicant; we as technical partner
IFC Climate-Smart Agriculture financing window	IFC	Loans + grants	Direct alignment with the new CSA Investment Plan launched June 2025	Engage IFC Resident Rep in Kigali
USAID Development Innovation Ventures (DIV)	USAID	USD 25-200k Stage 1, up to USD 5M Stage 3	Stage 1 fits us today	Rolling
World Bank Digital Development / D4Ag	World Bank	Variable	Component within larger programmes (Feed the Future, RDAP)	Via implementing partners
Mastercard Foundation Africa Growth Fund	Mastercard Foundation	USD 100k-2M	Youth employment + AgriTech angle works	Annual
UK FCDO Frontier Technology Hub	UK government	GBP 20-100k pilot funds	Climate + frontier tech	Rolling
EU's Global Gateway Africa-	EU	Large infrastructure	Long shot, partner-led	Via local primes

Programme	Funder	Amount	Best fit	Window
Europe		tickets via local		
Investment		partners		
Package				

7.3 Recommended grant strategy

Apply now (Month 1-3):

1. Hanga Pitchfest — fast money, builds credibility
2. RDAP Early Stage Financing — fast, local, low effort
3. GSMA Innovation Fund (whichever round is currently open)
4. Rwanda Green Fund (FONERWA) — frame as climate adaptation

Apply Month 4-9 (after we have a working pilot + first letter of intent from WASAC or similar): 5. Rwanda Innovation Fund / Angaza Capital — equity round 6. USAID DIV Stage 1 7. IFC engagement via the CSA Investment Plan 8. Mastercard Foundation Africa Growth Fund

Position parallel applications differently:

- For climate funders (GSMA Green, FONERWA, IFC): water + agriculture sustainability angle
- For digital inclusion funders (USAID DIV, Mastercard): rural farmer reach + women-led component
- For innovation funders (NCST, Hanga): the technology novelty + first-in-country positioning

7.4 Grant application narrative blocks (reusable text)

When writing applications, our story has four reusable narrative blocks. Mix-and-match per funder:

Block A — Climate / sustainability:

Rwanda loses an estimated 38% of its produced municipal water to leaks and unbilled consumption — over Rwf 9.9 B (USD 7.5M) annually, equivalent to a full year of operating budget for the national water utility. Our LoRaWAN-based smart-metering platform enables near-real-time leak detection at fraction of the cost of cellular alternatives, addressing both economic loss and the climate-adaptation imperative of efficient freshwater use.

Block B — Agricultural transformation:

Rwanda's Strategic Plan for Agricultural Transformation (PSTA5, MINAGRI 2025) aims to make 83,250 hectares climate-resilient and link 170,200 farmers to climate-smart financing. Our connectivity layer enables affordable IoT sensors (soil moisture, temperature, livestock tracking) at \$1.50/device/month — making digital agriculture viable for smallholder cooperatives, not just commercial farms.

Block C — Digital sovereignty / local ownership:

Our network is built on open-source ChirpStack with no foreign licensing fees, runs on infrastructure that can be hosted entirely within Rwanda, and creates local skilled employment in network engineering, RF deployment, and IoT solutions consulting. Unlike foreign vendors selling closed systems, we transfer capability — not just hardware.

Block D — Regional scalability:

The platform we operationalise in Rwanda is identical to what we would deploy in any East African Community country. A successful Rwandan deployment establishes the operational, regulatory, and commercial template for replicable expansion across the EAC, supporting integration objectives at the regional level.

8. Go-to-Market Plan

8.1 12-month sequence

Month	Milestone
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1-2	Procure 3 production gateways (EU868). Begin RURA type approval.
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2-3	Open conversations with WASAC, NAEB, City of Kigali innovation offices. Apply to Hanga Pitchfest + GSMA.
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3-4	Deploy gateways in pilot zones (1 indoor demo office + 2 outdoor sites). First "company" formed legally.
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4-6	Land first paid pilot. Target: 100-500 device deployment with one anchor customer.
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6-9	Operational pilot. Generate case study + data. Apply to Innovation Fund (equity round).
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9-12	Sign second + third customer. Expand gateway footprint to 10-15 in Kigali.
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8.2 Anchor pilot — the one that unlocks everything

Of all 31 organisations in Section 4, **the strategic anchor** is **WASAC**. Reasons:

- It validates the most-cited Rwanda IoT use case (water NRW)
- Government utility = reference customer for every subsequent agri/health/logistics conversation
- Their procurement is happening *now* (active RFPs for smart meter leasing in 2025-2026)
- Their CEO has publicly stated the strategic interest

Plan B if WASAC moves too slowly: anchor with NAEB or a Tea/Coffee export cooperative. Both have urgent quality-monitoring needs and faster procurement.

8.3 Sales motion

- Land via **direct relationships and demos**, not advertising
 - Every executive at the 6 Tier-A organisations should be approached individually within Month 1-3
 - For Tier B and C, run **1 vertical-specific demo day per quarter** (e.g. "smart agriculture demo at Kigali Innovation City" with 5 cooperative invitees)
 - Hire the first salesperson only after the WASAC anchor is signed
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9. Team & Capacity

(To be customised by founders.)

Capabilities required to operationalise:

- **Technical / RF / network engineering** — already on team (the person who built the proof of concept)
- **Business development & government affairs** — TBD; ideally an ex-WASAC, ex-RDB, or ex-MINICT senior who knows the procurement system
- **Field operations / installations** — TBD; can be hired locally
- **Finance / admin** — outsourced fractional for first 12 months

Advisory board to assemble:

- One senior figure from a Rwandan utility (water or electricity)
- One regional LoRaWAN operator (e.g. Upande Kenya CEO)
- One investor / VC familiar with African deeptech

- One agricultural sector veteran (NAEB or large cooperative)

10. Risks & Mitigations

Risk	Likelihood	Mitigation
RURA type approval delays > 3 months	Medium	Start the process Month 1; engage RURA briefing meeting Week 2; use pre-CE-certified gateway hardware to shorten the path
WASAC procurement timeline > 12 months	High	Multi-track Tier-A engagement (don't depend on one customer); use grant-funded pilots to build references in parallel
Foreign competitor enters Rwanda first	Medium	Speed: be operational within 6 months. Defensible moats: regulatory positioning, local relationships, in-language support
Hardware import delays (Rwanda customs / EU868 module availability)	Medium	Pre-order hardware Month 1; identify local distributor relationships
Cellular operators bundle NB-IoT cheaper than expected	Low-Medium	We retain advantages on battery life (5-10 yrs vs ~2 yrs), per-device cost, indoor penetration
Network reliability / SLA violations damage early reputation	High at start	Conservative SLAs (95% uptime initially, not 99.9%); transparent status page; on-call rotation by Month 6
Currency / forex (hardware in USD, revenue in RWF)	Medium	Price contracts in USD where customer can absorb; build forex buffer into Year 1 reserves

11. The Ask

This document is suitable for three purposes:

1. **Sent to a potential anchor customer** (e.g. WASAC, NAEB, City of Kigali) → use Sections 1, 2, and a customised version of 4.1 / 8.2
2. **Sent to a grant funder** → use Sections 1, 2, 3, 7, plus one of the narrative blocks in 7.4 customised

3. **Sent to an investor** → use Sections 1, 3, 5, 6, 8, 10 as the core; add detailed financial model as appendix

What we need now (Month 0):

- **USD 30k seed capital** to procure first 3 production gateways, complete RURA approval, and operate for 6 months while landing the anchor customer
 - **2-3 advisory board members** with relationships into WASAC / RDB / MINICT
 - **1 letter of intent** from any Tier-A organisation in Section 4.1 — even a non-binding LOI dramatically accelerates all subsequent fundraising
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Appendix A — Source data & references

1. WASAC NRW figures — RURA annual utility report; *The New Times* coverage of WASAC NRW losses
2. MINAGRI Climate-Smart Agriculture Investment Plan — launched 18 June 2025 with IFC
3. Itron-WASAC partnership — 160,000 smart meters deployed cumulatively
4. Rwanda Innovation Fund — managed by Angaza Capital, public information from MINICT
5. GSMA Innovation Fund — current funding rounds at gsma.com (verify dates at application time)
6. ChirpStack — open-source LoRaWAN network server, Apache 2.0 — chirpstack.io
7. Comparable operator Upande Limited (Kenya) — GSMA Innovation Fund alumnus

Appendix B — Glossary

- **LoRaWAN** — Long-Range Wide-Area Network. Open low-power radio standard for IoT.
 - **NRW** — Non-Revenue Water. Water that is produced but not billed.
 - **NaaS** — Network-as-a-Service. Connectivity sold as recurring service.
 - **NS** — Network Server (the ChirpStack software stack).
 - **RURA** — Rwanda Utilities Regulatory Authority.
 - **EU868** — The 863-870 MHz LoRaWAN frequency band used in Europe and most of Africa.
 - **ChirpStack** — Open-source LoRaWAN Network Server software, the foundation of our platform.
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